Dyeing Equipment



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Dyeing Set Up Equipment; Dyeing units

What it is used for

Dyeing equipment applies dyes to fibres, yarn, fabric, or garments under controlled process conditions to achieve the required colour and shade. Correct dye preparation, temperature and time control, chemical handling, and colour-quality checks help produce consistent results while reducing exposure and environmental risks.

Main parts

- Dye bath, vessel, or tank holds the dye liquor and textile material.
- Heating system controls bath temperature where fitted.
- Agitator or circulation pump helps distribute dye evenly.
- Fabric carrier, basket, reel, or loading frame holds the textile.
- Dosing system measures and introduces dye or auxiliaries where fitted.
- Temperature sensor and controller monitor process conditions.
- Drain and rinse connections remove spent liquor and support cleaning.
- Control panel manages operating parameters.
- Exhaust or ventilation system removes vapours or aerosols where required.

Before you start

1. Only trained personnel should operate dyeing equipment and handle dye chemicals.
2. Read the machine instructions and confirm the approved dye recipe, fabric type, liquor ratio, temperature, and processing time.
3. Inspect the tank, seals, valves, agitator, pumps, heating system, guards, and controls for damage or leaks.
4. Verify that ventilation and any required chemical extraction are operating properly.
5. Check dye and auxiliary chemical labels, safety data sheets, compatibility, and required protective equipment.
6. Prepare the dye solution using the approved recipe and chemical-handling procedure; never mix unknown substances.
7. Wear suitable chemical-resistant gloves, eye protection, protective clothing, and respiratory protection when required by the risk assessment.

How to use

1. Confirm that the vessel, textile loading arrangement, and drain route are clean and ready for use.
2. Weigh and prepare the textile material according to the approved batch instructions.

3. Measure water and chemicals accurately according to the authorised dye recipe.
4. Fill the vessel to the specified level and add dye and auxiliaries in the prescribed sequence.
5. Load the textile evenly and avoid overloading or obstructing circulation.
6. Set the temperature, time, agitation, and other process parameters according to the recipe.
7. Start the dyeing cycle and monitor temperature, circulation, chemical dosing, and machine operation.
8. Take process samples only using the approved sampling method and assess shade or process progress where required.
9. Stop the process if leakage, overheating, abnormal vibration, or a chemical handling incident occurs.
10. Complete dyeing and carry out rinsing, neutralisation, or after-treatment according to the approved process.
11. Drain the bath through the authorised waste-treatment route and never discharge untreated chemicals into an unauthorised drain.
12. Unload the textile safely, inspect colour uniformity, and record batch details and deviations.
13. Clean the vessel and relevant components according to the chemical and machine instructions.

⚠ Safety

- Follow chemical safety data sheets and never mix incompatible dyes, acids, alkalis, or auxiliaries.
- Avoid skin and eye contact with dye liquor and chemical concentrates.
- Never open a hot or pressurised vessel until it is safe and pressure has been released as designed.
- Keep hands away from agitators, rollers, pumps, and other moving parts.
- Isolate energy sources before cleaning, clearing blockages, or maintenance.
- Treat dye effluent according to local environmental rules and the facility's approved wastewater procedure.

Care & storage

- Clean the vessel, filters, dosing lines, and textile-contact components after each batch as required.
- Inspect valves, seals, hoses, heating elements, sensors, and pumps for leaks or wear.
- Calibrate dosing and temperature controls at the prescribed intervals.
- Remove dye deposits using approved cleaning methods compatible with equipment materials.
- Report persistent shade variation, flow problems, or control faults and arrange authorised maintenance.

Common mistakes to avoid

- Using the wrong dye recipe, liquor ratio, temperature, or processing time.
- Adding chemicals in the wrong sequence or measuring them inaccurately.
- Overloading the vessel and causing uneven dye circulation.
- Discharging untreated dye liquor or cleaning chemicals into unauthorised drains.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.